higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATED GC

NATIONAL CERTIFICATE

NOVEMBER EXAMINATION

INDUSTRIAL INSTRUMENTS N5

2 December 2015

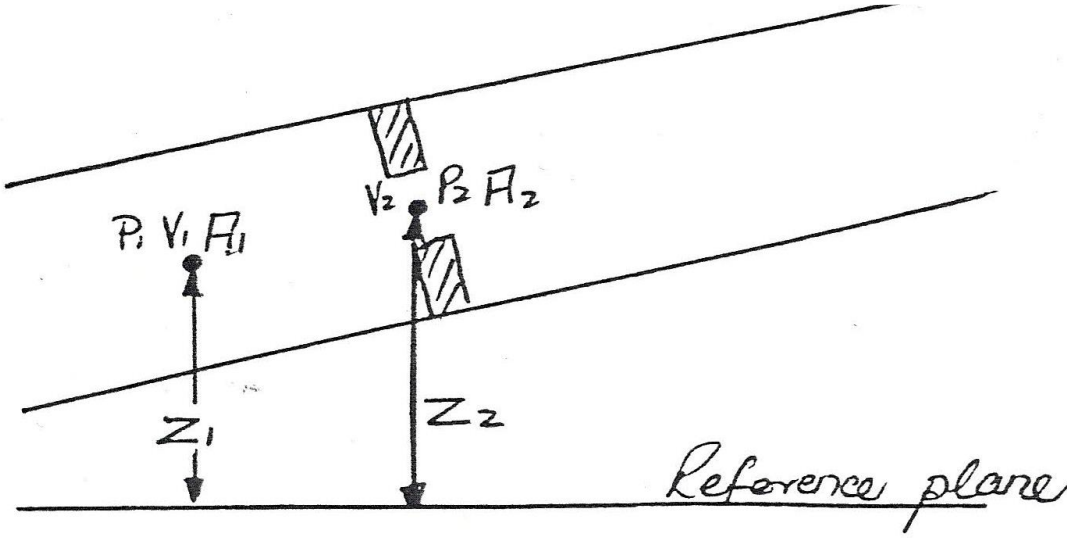
This marking guideline consists of 11 pages.

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SECTION A

QUESTION 1

1.1



Potential energy = mgh
Kinetic energy = ½ mV²
Pressure energy = PV
Internal energy = I (5)

1.2 PE + KE + P V + I = PE + KE + P V + I (½)

$m.gz + \frac{mV^2}{2} + P V + I = mgz + \frac{mV^2}{2} + P V + I$ (½)

If temperative is constant I = I for liquids only (½)

$V = V = \frac{M}{d} = \frac{M}{d} = \frac{M}{d}$ density = mass x volume (½)

$mgz + mV^2 + \frac{P\ m}{d} = mgz + mV^2 + \frac{P\ m}{d}$

divide the equation by m.g.

$Z + \frac{V^2}{2g} + \frac{P}{dg} = Z + \frac{V^2}{2g} + \frac{P}{gd}$ (½)

$\text{thus } (Z + \frac{P}{gd}) - (Z + \frac{P}{gd}) = \frac{V_1^2}{2g} - \frac{V_2^2}{2g}$ (½)

$h = \frac{V_1^2}{2g} - \frac{V_2^2}{2g}$ (½)

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$2gh = \frac{V_1^2}{} - \frac{V_2^2}{}$ (½)

$Q = A_1 V_2 = A_1 V_2$

$V_1 = \frac{A_2 V_2}{A_1}$ (½)

$2gh = \frac{V_2^2 - A_2^2 V_1^2}{A_1^2}$ (½)

the ratio A/A is known as the area ratio 'm' (½)

$2gh = V_2^2 - V_1^2 m^2$ (½)

$2gh = V_2^2 (1 - m^2)$ (½)

$2gh \cdot \frac{1}{1-m^2} = V_2^2$ (½)

$V = 2gh.1/1-m^2$ (½)

The term $1/\sqrt{1-m^2}$ is known as the velocity of approach factor and is represented by 'E' (½)

$V = E. \sqrt{2gh}$ (½)

thus $Q = A_1 V_1 = A_2 V_2$ (½)

$V = \frac{Q}{A_1}$ (½)

$\frac{Q}{A} = E. \sqrt{2gh}$ (½)

$Q = A E \sqrt{2gh}$ (½)

But 'C' is known as the discharge coefficient and equals the actual volume flowing divided by the theoretical volume. (½) (½)

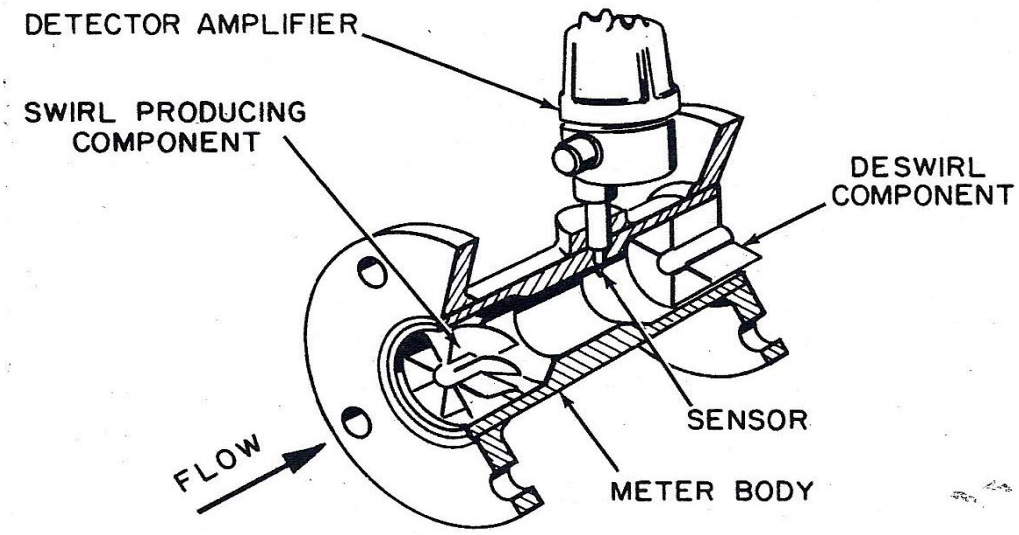
$Q = C.A .E. \sqrt{2gh}$ (½)

(12)
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QUESTION 2

2.1



The vortex centre is a region of higher velocity than the rest of the fluid profile. The sensor, a thermistor, detects the velocity change.

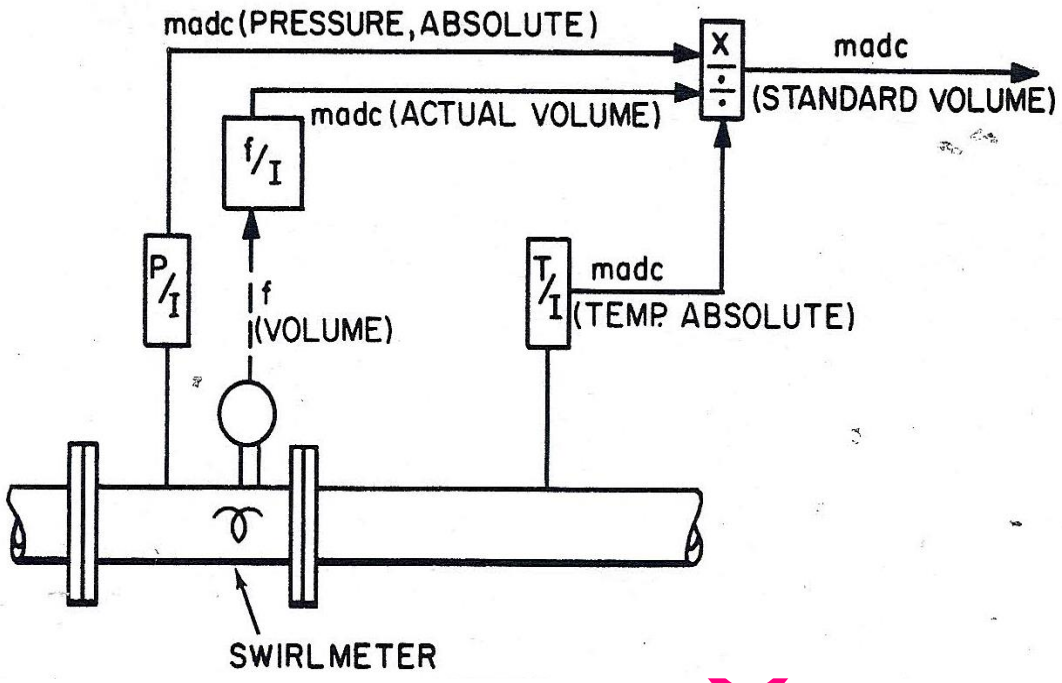
The frequency with which the velocity changes are occurring is proportional to flow-rate.

The thermistor senses the change in temperature resulting from the increased cooling due to the passing region of increased velocity.
The thermistor resistance changes with temperature so each passing of the region of higher velocity results in a resistance change. The thermistor has a constant current applied to it, so the resistance change is detected as a voltage change which is subsequently amplified and filtered. The output signal is a 15 volt peak-to-peak square wave, the frequency of which varies with flow-rate.

Just before the outlet of the meter, the fluid passes through the de-swirl component. This component is intended to isolate the measurement portion of the swirlmeter from downstream piping effects which could propagate opposite to the direction of flow and impair the development of the processing vortex. (10)

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2.2



(4)
[14]

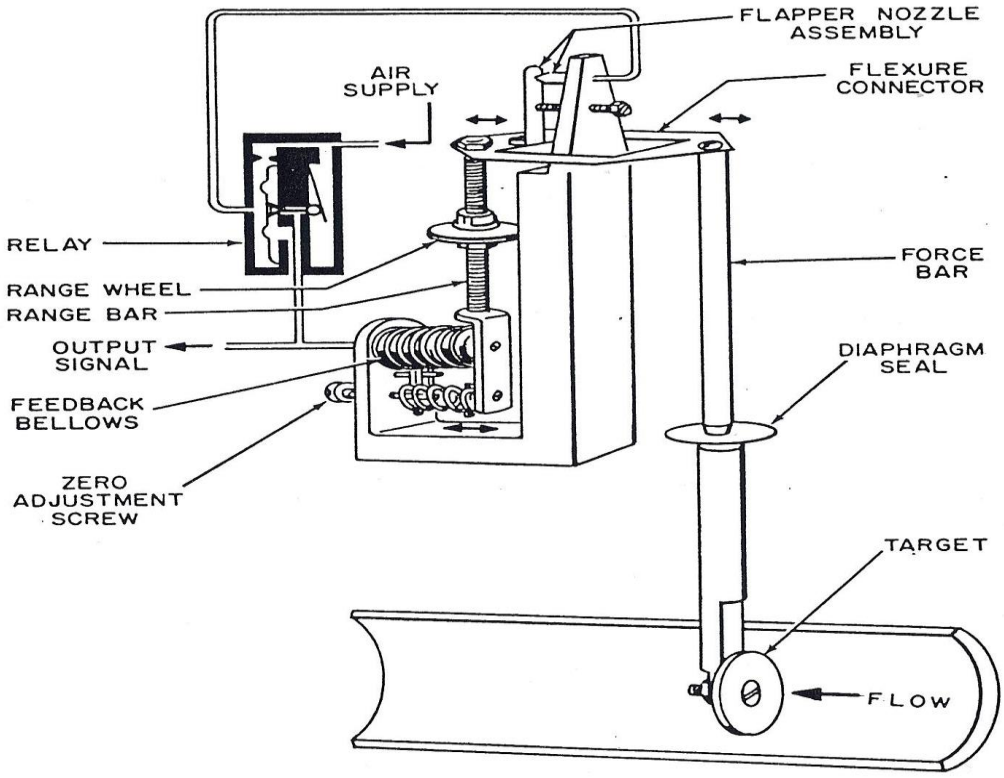
TOTAL SECTION A: 31

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SECTION B

QUESTION 3

3.1



The target meter combines in a single unit an annular orifice and a force-balance transducer. Output is either electric or pneumatic signal proportional to the flow rate.

The target is formed by a circular disc supported in the centre of a tubular. Flow through the open ring between disc and tube develops a force on the disc proportional to the square of the flow.

The disc is mounted on a rod passing out through a flexible seal (diaphragm seal). The force on the disc is measured from the rod outside the seal using a standard force-balance transducer mounted on the flow tube (flapper, nozzle assembly). (6)

3.2 Advantages

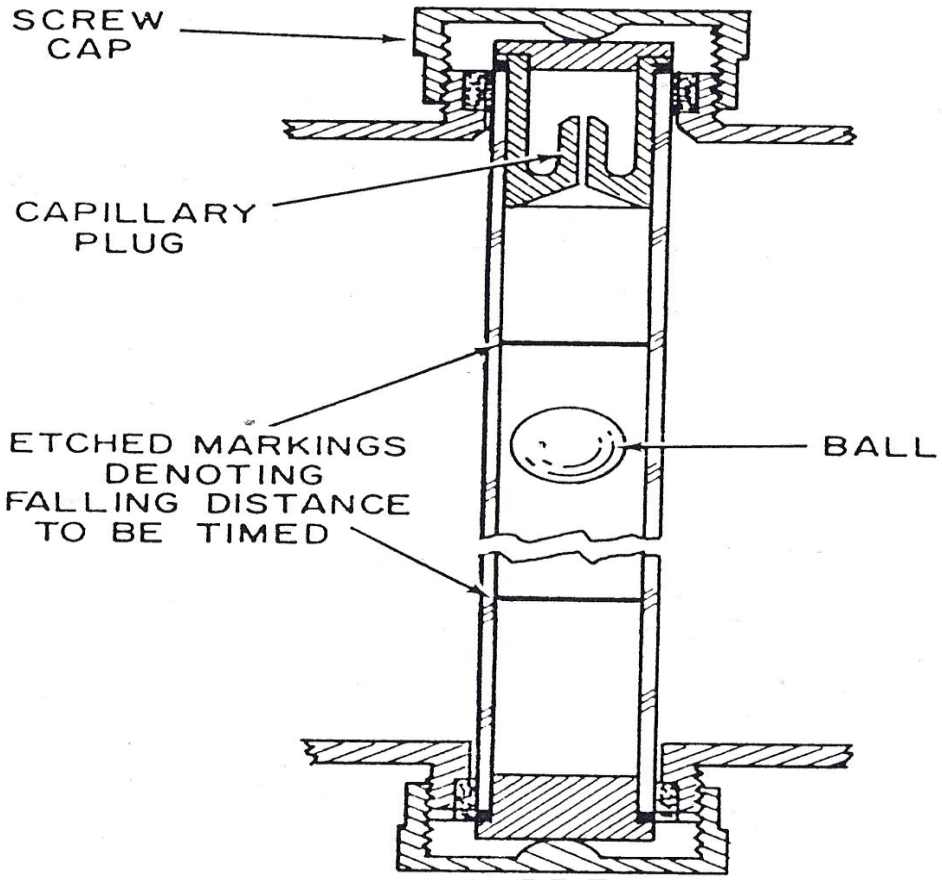
- Can measure flow of very high viscosity fluids, eg. tar, oils and slurries. Can be used in applications where other meters were unsuccessful.
- Can measure pressures up to 100 bar
- Repeatability
- Accurate (Any 2 × 1) (2)

Disadvantages

- Expensive
- In-line connection (2)

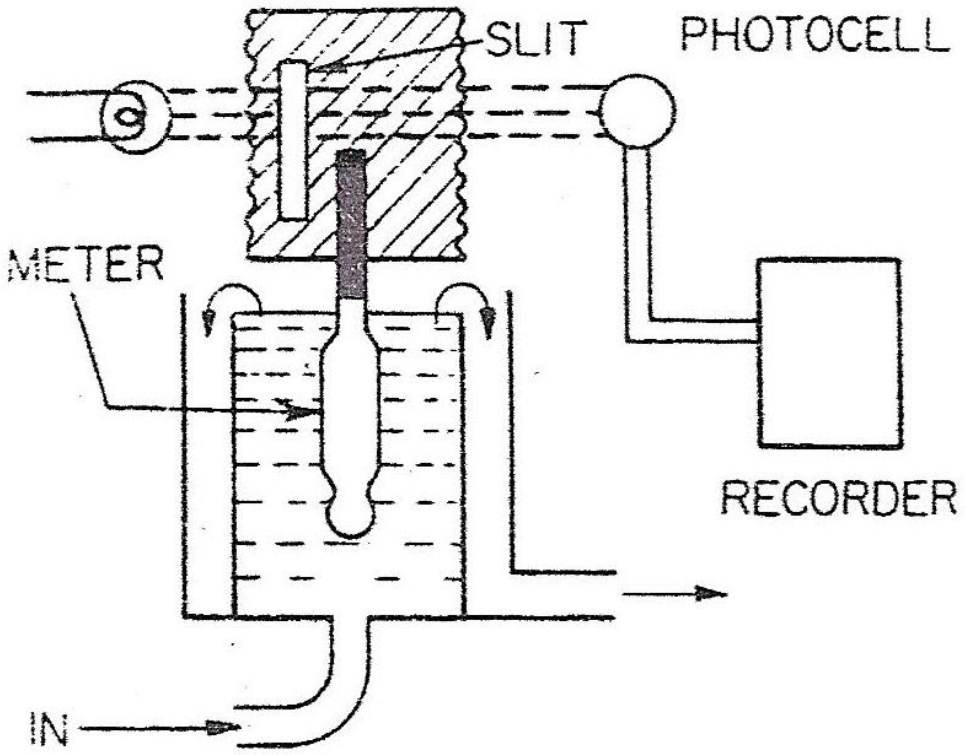
[10]

QUESTION 4


SCREW CAP
CAPILLARY PLUG
ETCHED MARKINGS DENOTING FALLING DISTANCE TO BE TIMED
BALL

[4]

VRAAG 5



A hydrometer with a dull stem is placed in a continuous flow vessel. The liquid must be kept at the same level. The liquid with a specific density will move the hydrometer to a certain level. The blacked-out stem will then block off a certain amount of light from the light source. These light rays will be picked up by a feed cell which in turn will send a signal to the recorder. The amount of light received by the fotocell is directly proportional to the density of the liquid. **[10]**

QUESTION 6

6.1 To provide automatic temperature compensation, requires a property of the zero shift and slope shift voltages to be equal and opposite. This part on the pH scale is known as the isopotential point and at this point the electrodes are self-compensating. (2)

6.2 The number of hydroxyl ions exceeds the number of hydrogen ions and the solution is basic. (3)

6.3
- It must have resistance to corrosion.
- It must have strength against thermal shock.
- It must have strength against mechanical shock.
- It must be impermeable to water vapour at high temperatures. (4)

[9]

TOTAL SECTION B: 33

SECTION C

QUESTION 7

7.1
a) All four bellows will have the same pressure.
b) The motion pin will be on its true centerline.
c) There will be no control actions generated. (3)

7.2
a) The output will increase by ± 10 kPa if the gain is direct.
b) Should the gain be indirect the output will decrease by ± 10 kPa.
c) This would be due to the pressure increasing in the process bellows while the pressures in the other bellows stay constant.
d) The motion pin will move off its centerline causing the flapper to move either towards or away from the nozzle.
e) This should cause the nozzle feedback pressure to either increase or decrease thus causing the output to change accordingly. (4)

7.3 100%

$$Pb = \frac{100}{Gain}$$

(1)

7.4 By swiveling the gain adjustment dial from direct to reverse. (1)

7.5 When the integral action is marked in repeats per minute it indicates that the integral will repeat the proportional band that many times per minute.

When marked in minutes per repeat it indicates the time to repeat the proportional band. (1)

7.6 The derivative action-time is the time taken for the proportional action to repeat the derivative action for a ramp change. Most controllers do not produce the true theoretical outputs due to interaction between the control actions and are therefore called interacting controllers. (2)

7.7 Most controllers do not produce the true theoretical output. (2)

7.8 Advantages of a live signal:

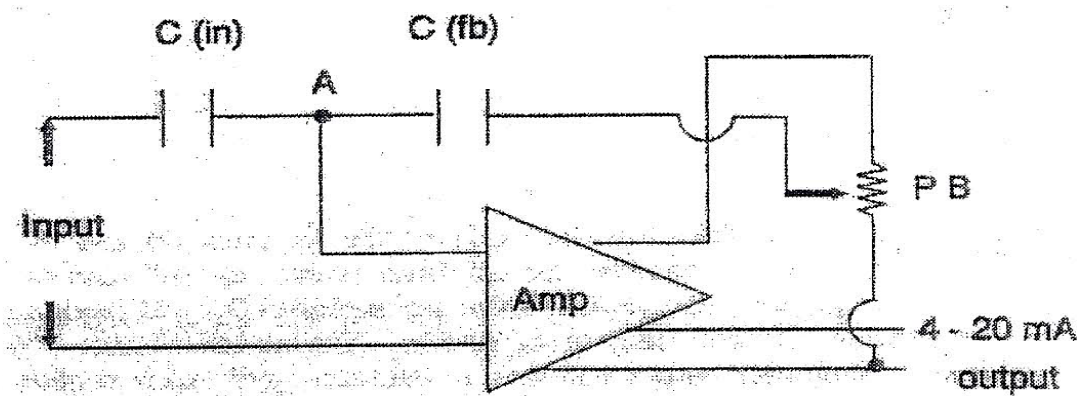
a) Difference between power on and power off
b) See movement from zero in both directions
c) Two-wire system (: - no high voltage in hazardous areas) (3)

Advantage of a zero-based signal:
Mathematical functions are easy, as they do not have to be biased. (1)
[18]

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QUESTION 8

8.1



Operation:

- Cin and Cfb are equal in value. (2)

There are two outputs from the amplifier:

- A current signal ranging from 4 to 20 mA for control purposes.
- A voltage signal for feedback purposes (1)

These two are proportional to one another.

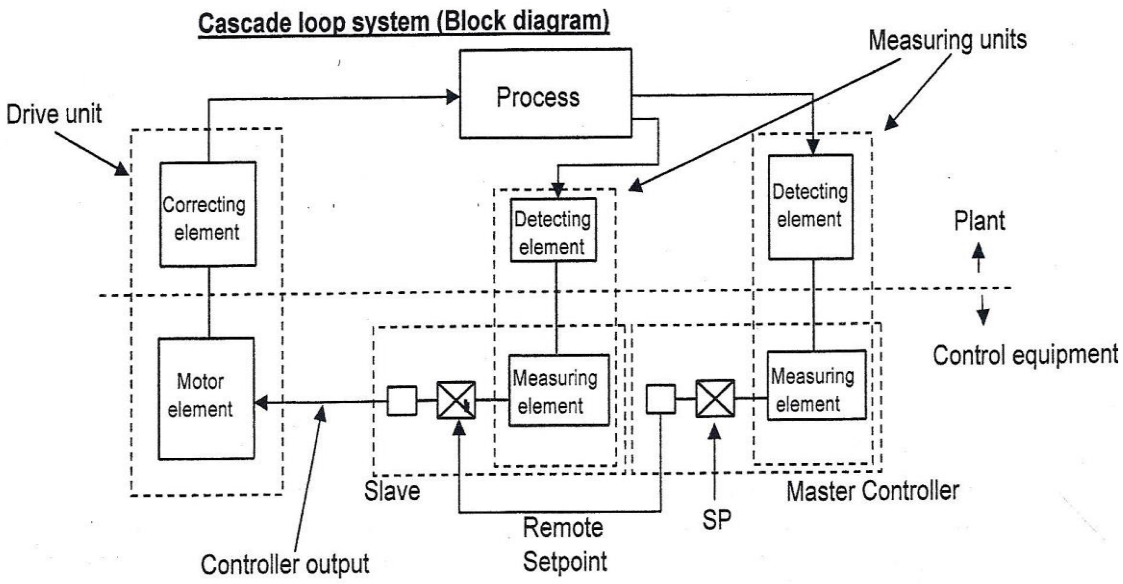
With no input signal, the feedback voltage is zero and the potential at A, relative to earth, is zero. There will be an output signal, however, and this could conveniently be 12 mA, the half-way value between 4 and 20 mA. (2)

If an error signal is impressed on the input, capacitor Cin is charged and a potential appears at point A, forming an input signal to the amplifier. An output current (for correction) and feedback voltage is generated to Cfb to counteract this and the potential at A returns to zero. This is, of course, the same conditions as we had for the original signal output of 12 mA. (2)

For as long as there is a sustained input signal to the amplifier, there is an equally sustained feedback voltage generated, and if this changes, so does the output current in proportion. (1)

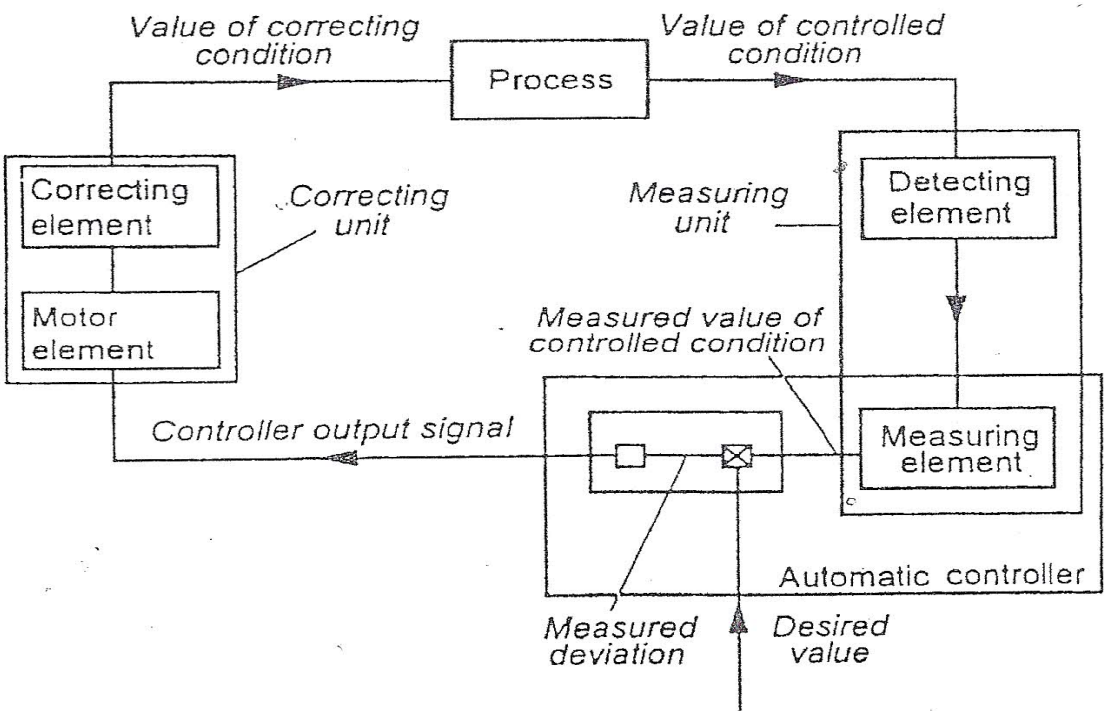
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8.2



(5)

8.3



A combination of control units in which the process variable is measured and compared with the desired value (or set point).
If the measured value differs from the desired value, a corrective signal is sent back to the final control element to bring the controlled variable to the proper value. (5)

[18]

TOTAL SECTION C: 36
GRAND TOTAL: 100

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